

Pankaj Singh — Curriculum Vitae

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Education

University of South Carolina

Ph.D. Mathematics

Advisor: Dr. Alexander Duncan

Central Michigan University

M.S. Mathematics

Thesis: *Connectivity of the Tropical Double Ramification Cycle*

Advisor: Dr. Dmitry Zakharov

IIT Guwahati

M.Sc. Mathematics and Computing

Mahatma Gandhi Kashi Vidyapeeth

B.Sc. Mathematics and Statistics

Columbia, SC

Aug '20 – Expected 2026

Mt. Pleasant, MI

Aug 2019 – Jun 2020

Guwahati, India

Varanasi, India

Research & Interests

Current Research:

- Classification of higher-dimensional tori and toric varieties over arbitrary fields
- Ideals from powers of general linear forms and the Weak Lefschetz Property

Research Interests: Algebraic Geometry, Galois Cohomology, Brauer Groups, Commutative and Noncommutative Algebra, Representation Theory, Mackey Functors

Publications & Preprints

- **On the Weak Lefschetz Property for Ideals Generated by Powers of General Linear Forms**
(with Matthew Booth and Adela Vraciu), To appear in *Journal of Commutative Algebra*.
<https://arxiv.org/abs/2410.22542>
- **Classifying Torsors of Tori via Brauer Groups**
(with Alexander Duncan), Submitted.
<https://arxiv.org/abs/2505.10386>
- **Classifying Torsors of Tori of del Pezzo variety**
In progress

University Teaching Experience

Instructor of Record — University of South Carolina.....

- MATH 174 — Discrete Math for Computer Science
(logic & introduction to proofs; 50 students) Spring 2025
- MATH 111I — Intensive College Algebra (60 students) Fall 2024
- MATH 122 — Calculus for BA & Social Sciences (52 students) Spring 2024
- MATH 111 — College Algebra (50 students) Fall 2023
- MATH 111I — Intensive College Algebra (60 students) Spring 2023
- MATH 170 — Finite Mathematics (45 students) Fall 2022
- MATH 122 — Calculus for BA & Social Sciences (52 students) Summer 2022

- MATH 142 — Calculus II (35 students) Summer 2021
- Instructor of Record — Central Michigan University.....
- MATH 105 — College Algebra (40 students) Spring 2020
- MATH 105 — College Algebra (42 students) Fall 2019
- Teaching Assistant — University of South Carolina.....
- MATH 141 — Calculus I (50 students) Fall 2020 & Fall 2021

Awards & Scholarships

2024: Two Thumbs Up Award, University of South Carolina — Recognized by Student Disability Services for contributions to an inclusive and supportive learning environment.

2025: Thank-a-Teacher Certificate, University of South Carolina — Awarded by the Center for Teaching Excellence for making a positive impact on students' learning.

Talks & Presentations

Upcoming Talks.....

- International Congress of Mathematicians, Pennsylvania Convention Center, Philadelphia 2026

Invited Talks.....

- AWM Research Symposium, University of Wisconsin–Madison 2025
- Algebra, Geometry, and Number Theory Seminar, University of South Carolina 2025
- Algebra Seminar, University of Georgia 2025

Contributed & Lightning Talks.....

- Summer Research Institute in Algebraic Geometry (Flash Talk), Colorado State University 2025
- Commutative Algebra & Algebraic Geometry Seminar, University of Minnesota 2025
- Georgia Algebraic Geometry Symposium (Lightning Talk), University of Georgia 2025
- Algebraic Geometry Northeastern Series (AGNES) (Lightning Talk), Dartmouth College 2024
- AMS Fall Western Sectional Meeting, Special Session on Finite Groups, Their Representations, and Related Structures, University of California, Riverside 2024
- Palmetto Number Theory Series (PANTS XXXVIII), Wake Forest University 2024
- Motivic Homotopy, K-Theory, and Modular Representations Conference (Lightning Talk), University of Southern California 2024

Poster Presentations.....

- AWM Research Symposium, University of Wisconsin–Madison 2025
 - Algebraic Geometry Northeastern Series (AGNES), Dartmouth College 2024
- Topic: "On the Weak Lefschetz Property for Ideals Generated by Powers of General Linear Forms"*

Conferences & Workshops

2025.....

- Professional Development Workshop for Early-Career Commutative Algebraists, University of Nebraska–Lincoln 2025
- Summer Research Institute in Algebraic Geometry, Colorado State University 2025
- AWM Research Symposium, University of Wisconsin–Madison 2025
- Georgia Algebraic Geometry Symposium, University of Georgia 2025
- The Legacy of John Tate and Beyond, Harvard University 2025

Past Selected Conferences & Workshops.....

- Algebraic Geometry Northeastern Series (AGNES), Dartmouth College 2024
- AMS Fall Western Sectional Meeting, University of California, Riverside 2024

- Motivic Homotopy, K-Theory, and Modular Representations Conference, University of Southern California 2024
- Workshop on Brauer Groups and Derived Categories, Northwestern University 2023
- Arizona Winter School (PAWS), University of Arizona 2023
- Arithmetic Geometry and Algebraic Groups, University of Virginia 2023
- Algebraic Geometry Northeastern Series (AGNES) 2022 (Virtual), Rutgers University 2022
- Combinatorial Algebraic Geometry: Tropical and Real, ICTS, India 2022

Selected Teaching Seminars & Workshops Attended

- Entering Mentoring: Introduction to Mentor Training 2024
- Having Constructive Classroom Conversations in Challenging Times (ACUE) 2024
- Setting Classroom Expectations and Creating a Sense of Belonging (Center for Teaching Excellence, USC) 2024
- Reimagining Assessments with Generative AI 2024
- AMS Fall 2024 Mathematics Education, Special Session on Understanding Student Thinking 2024
- Redesigning an Upper-Division Math Course for Future Secondary Teachers 2024
- The Impact of a Redesigned Differential Equations Course on Prospective Teachers 2024

Judging & Reviewing

- Reviewer, Undergraduate Poster Presentations at Discover USC, Columbia Metropolitan Convention Center Spring 2023
- Reviewer, South Carolina Junior Science and Humanities Symposium (JSHS) 2024–2025
- Judge, USC Region II Science and Engineering Fair Spring 2025

Outreach & Mentorship

- Mentor, Online Teaching Support for New Graduate Instructors, University of South Carolina Spring 2024
- Volunteer, High School Math Contest, University of South Carolina Spring 2025
- Panelist, Graduate Teaching Assistant Orientation, University of South Carolina Fall 2022

Technical Skills

- **Statistical Software:** SAS (project-based learning) — data cleaning, analysis, visualization
- **Programming:** Python, SageMath — matrix computations, module and ring theoretic calculations
- **Computer Algebra Systems:** Macaulay2 — Gröbner basis, Hilbert functions, and exact zero-divisor pair computations
- **Machine Learning:** Applied regression, gradient descent, classification models, and overfitting mitigation
- **Typesetting:** $\text{\LaTeX}$ for research articles, teaching, and presentations

Selected Professional Service & Development

- Co-organizer of the 2025 Fall Eastern Virtual Sectional Meeting (with Swati and Shreya Sharma): AMS Special Session “Unveiling Connections: Number Theory Meets Algebraic Geometry” 2025 (Fall)
- Co-organizer (with Anirban Bhaduri and Matthew Booth), Reading Seminar: Toric Geometry in Mirror Symmetry, University of South Carolina Summer 2025
- Co-organizer (with Bailey Heath), Graduate Colloquium, University of South Carolina Fall 2023–Spring 2024
- Co-organizer (with Anirban Bhaduri), Algebraic Geometry, Commutative Algebra, and Number Theory Student Seminar, University of South Carolina Fall 2023–Spring 2024
- Participant, Preparing for Future Faculty Program, Center for Teaching Excellence, University of South Carolina Spring 2024
- Participant, SHARPGrads Program, University Libraries, University of South Carolina Spring 2024